

Swimming Facility Inspection Outcomes in Toronto

Vijval Rallan

2026-09-28

1 Introduction

2 Data

2.1 SwimSafe Inspection Data

The SwimSafe dataset contains records from inspections of swimming facilities in Toronto. The raw dataset contains multiple records for some inspections because a single inspection can include multiple observations or recorded infractions.

For this analysis, an inspection is defined as one establishment, one facility type, and one inspection date. Records were therefore grouped using these three variables and aggregated to create an inspection-level dataset. This resulted in 9,967 inspections from 17,891 raw records.

The analysis focuses on four facility types: Indoor Pool, Outdoor Pool, Spa Indoor, and Spa Outdoor. Inspection outcomes are recorded in the **insStatus** variable as Pass, Conditional Pass, or Closed. Individual recorded infractions are categorized in **infType** as Minor, Significant, or Crucial, while **infCategory** describes the category of the recorded infraction.

Table 1: Example observations from the SwimSafe inspection data

facilityName	type	insStatus	insDate	infType	infCategory
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-06-04	None	None
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-03-11	None	None
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-01-27	M - Minor	4. MAINTENANCE,
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-01-27	M - Minor	5. CIRCULATION
YORK RECREATION CENTRE	Indoor Pool	Pass	2025-11-07	None	None

Table 2: Number of inspections by facility type

type	inspections
Indoor Pool	5265
Outdoor Pool	1463
Spa Indoor	3041
Spa Outdoor	198

2.2 Inspection-level data

The analysis dataset contains 9,967 inspections across four facility types. Indoor pools account for the largest number of inspections, while outdoor spas account for the fewest. The difference in the number of observations across facility types is important when interpreting comparisons, particularly for outdoor spas.

3 Results

3.1 Inspection outcomes

Figure 1 shows the distribution of inspection outcomes across the four facility types. Indoor pools have the highest proportion of inspections recorded as Pass, at 86.9%, while Outdoor Pools and Spa Outdoor facilities have lower proportions of Pass inspections, at 69.9% and 68.7%, respectively. The proportion of inspections that were not recorded as Pass was therefore higher for outdoor facility types than for Indoor Pools. Spa Indoor facilities fall between these groups, with 79.7% of inspections recorded as Pass.

3.2 Recorded infraction severity

Figure 2 shows that the composition of recorded infractions also differs across facility types. Indoor Pools have the largest share of recorded infractions classified as Minor, at 68.9%, whereas Spa Outdoor facilities have the smallest share, at 49.1%. The combined share of Significant and Crucial infractions ranges from 31.1% for Indoor Pools to 50.9% for Spa Outdoor facilities. These percentages describe the distribution of recorded infractions within each facility type; they do not represent the percentage of inspections with an infraction of a particular type.

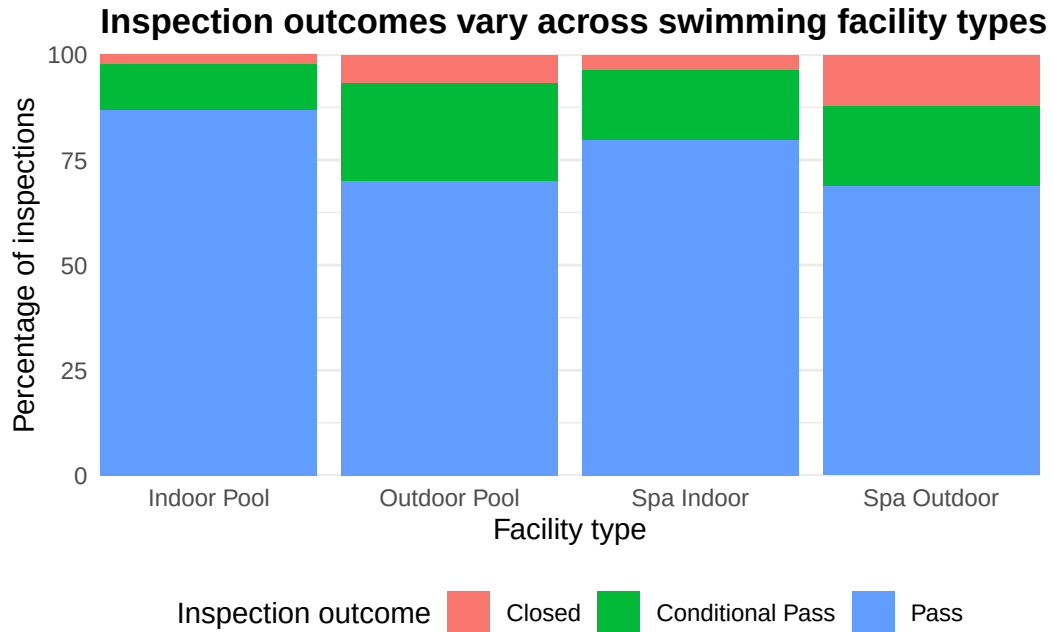


Figure 1: Inspection outcomes by swimming facility type.

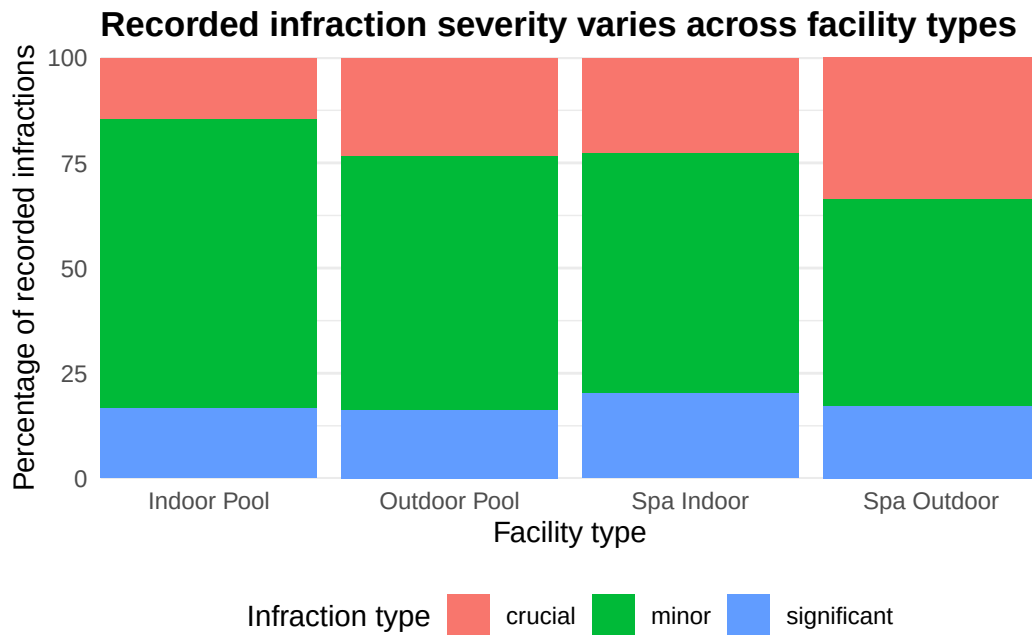


Figure 2: Distribution of recorded infraction types by swimming facility type.

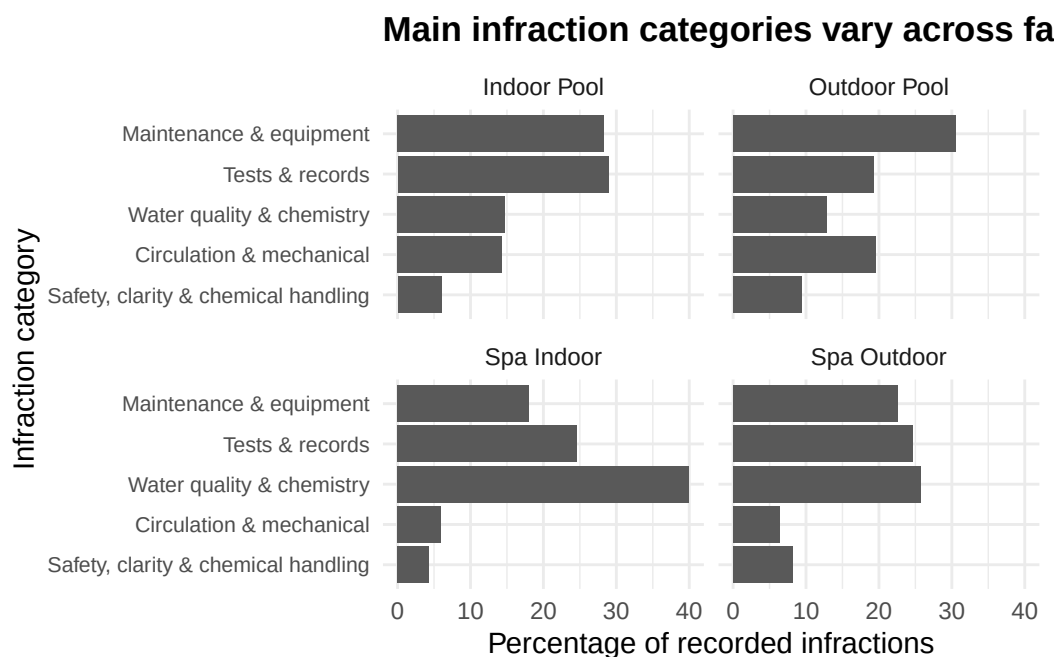


Figure 3: Main recorded infraction categories by swimming facility type.

3.3 Infraction categories

Figure 3 shows differences in the categories of recorded infractions across facility types. For Indoor Pools, Tests & Records and Maintenance & Equipment account for the largest shares of recorded infractions. Outdoor Pools have a relatively large share of Maintenance & Equipment and Circulation & Mechanical infractions. In contrast, Water Quality & Chemistry is the largest category for Spa Indoor facilities. These patterns indicate that the types of issues recorded during inspections are not distributed uniformly across facility types.

3.4 Common specific deficiencies

The specific deficiencies provide further detail on the broader categories shown in Figure 3. Among Indoor Pools, maintenance of pool and deck equipment and first aid kits is the most frequently recorded specific deficiency, followed by pool alkalinity and pool pH. For Spa Indoor facilities, spa alkalinity and spa pH are the most frequently recorded deficiencies. The most common deficiencies therefore differ across facility types, although some relate to water quality and maintenance across multiple types.

Table 3: Three most frequently recorded specific deficiencies by facility type.

type	defDesc
Indoor Pool	Maintenance of pool/spa deck equipment and first aid kit
Indoor Pool	Fail to ensure alkalinity of public pool water maintained in range of 80 - 120 ppm. Section 7 (8)
Indoor Pool	FAIL TO ENSURE PH VALUE OF PUBLIC POOL WATER MAINTAINED IN RANGE OF 7.2 - 7.8
Outdoor Pool	Maintenance of pool/spa deck equipment and first aid kit
Outdoor Pool	Fail to ensure all components of pool maintained in proper working order. Section 6 (6) (a)
Outdoor Pool	FAIL TO ENSURE PH VALUE OF PUBLIC POOL WATER MAINTAINED IN RANGE OF 7.2 - 7.8
Spa Indoor	Fail to ensure alkalinity of public spa water maintained in range of 80 - 120 ppm. Section 7 (8)
Spa Indoor	FAIL TO ENSURE PH VALUE OF PUBLIC SPA WATER MAINTAINED IN RANGE OF 7.2 - 7.8
Spa Indoor	Fail to ensure total bromine of public spa water maintained between 5 - 10 ppm. Section 7 (8)
Spa Outdoor	FAIL TO ENSURE PH VALUE OF PUBLIC SPA WATER MAINTAINED IN RANGE OF 7.2 - 7.8
Spa Outdoor	Health hazard conditions
Spa Outdoor	Fail to ensure residual of free available chlorine of public spa water maintained between 5 - 10 ppm. Section 7 (8)

4 Discussion

4.1 Main findings

This analysis shows that inspection outcomes and recorded infraction patterns vary across Toronto’s swimming facility types. Indoor Pools had the highest proportion of inspections recorded as Pass, while Outdoor Pools and Spa Outdoor facilities had higher proportions of non-Pass outcomes. The composition of recorded infractions also differed across facility types: Indoor Pools had larger shares of infractions related to Tests & Records and Maintenance & Equipment, Outdoor Pools had relatively larger shares related to Maintenance & Equipment and Circulation & Mechanical Systems, and Spa Indoor facilities had a particularly large share related to Water Quality & Chemistry.

Taken together, these results suggest that swimming facility types have distinct inspection profiles. The differences are visible both in the overall inspection outcome and in the categories and types of infractions recorded during inspections. However, these descriptive patterns should not be interpreted as evidence that facility type itself causes differences in inspection outcomes or that one facility type is inherently safer or less safe than another.

4.2 Limitations

Several limitations should be considered when interpreting these results. First, the analysis is descriptive and does not establish causal relationships between facility type and inspection outcomes or infractions. Second, the number of inspections differs substantially across facility types, with Spa Outdoor facilities having considerably fewer observations than Indoor Pools. Third, the analysis reflects conditions recorded during inspections and therefore does not necessarily represent conditions at a facility outside the inspection period. Finally, the categories Minor, Significant, and Crucial are used as recorded in the dataset; this analysis does not assign additional definitions or meanings to these categories beyond the available metadata.

5 Conclusion

This analysis examined how inspection outcomes and recorded infraction patterns vary across swimming facility types in Toronto. The results show differences in both the distribution of inspection outcomes and the composition of recorded infractions, with outdoor facility types having higher observed non-Pass rates than Indoor Pools and different distributions of infraction categories across facility types. The findings provide a descriptive view of the patterns present in Toronto's SwimSafe inspection records and illustrate how open administrative data can be used to examine differences across types of public facilities.